

Sport Climbing During Pregnancy

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Isabelle Schöffl in the first ascend of "Dreamcatcher",
Na Pha Daeng, Laos, Photo Volker Schöffl

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With its low risk of trauma, broad variety of moves, freedom to choose the individual load intensity, and technical difficulty, sport climbing developed into a popular recreational and competition sport [1]. Many years ago, climbing during pregnancy was considered to be a “no go” as data was sparse. The sport was also subjectively perceived as being a high-risk activity, even though significant objective data demonstrated its safety and low risk of injury and trauma compared with other sports (e.g., contact sports). The majority of medical professionals do not have an appreciation or understanding of the risks involved in the sport, and how they can be managed. The aim here is to provide information about sport climbing during pregnancy to care for the mother and fetus without being too restrictive.

It is now broadly encouraged that women with uncomplicated pregnancies should be encouraged to continue their aerobic and strength-conditioning exercises before, during, and after pregnancy, albeit with some modification due to anatomic and physiologic changes, type of exercise, and fetal requirements [2].

There is considerable medical literature about sports and pregnancy—but not including the different disciplines of climbing. However, even with this wealth of data in other sports, the quality of studies is often limited and subject to a more or less significant bias [3]. This includes the effect of physical exercise of moderate to vigorous intensity and frequency in the different trimesters as additional independent variables [4].

Some exercises that are recognized as safe for women with uncomplicated pregnancies who participated in such exercise previously in a meta-analysis include swimming, yoga (modified), pilates (modified), and strength training in consultation with an obstetric care provider [4]. This is also true for moderate strength training which shows some biomechanical similarities to sports climbing [5–9]. There are few maternal medical conditions where aerobic exercise is absolutely contraindicated [2]. However, sports featuring sudden deceleration or risk of collision are to be avoided—e.g., downhill ski racing, ice hockey, equestrian activities, pole vaulting [10].

With respect to sport climbing exercise, it should be differentiated whether the pregnant woman is an experienced climber or not. The former can continue climbing with some minor constraints, and the latter should be advised not to start climbing until the pregnancy has been completed. This strategy is recommended for any sport with complex strong and supple moves, (potential) risk of trauma, requirement for experience in using specialist equipment and techniques for safety (e.g., belaying and abseiling), and increased intra-abdominal pressure or which shows characteristics of competition.

Sports to be avoided for physiological reasons include diving as the fetus is not protected from decompression problems which risks malformation and gas embolism after decompression disease. Sport climbing or training at high altitude also presents physiological challenges, a topic discussed later.

All recommendations provided below are for active and experienced sport climbing women with uncomplicated pregnancies. The pathbreaking principle is the question of how the woman feels. In the case of a high-risk pregnancy, any advice depends significantly on the individual risk and also on the mentality of the woman and her need for security. Such advice would fall beyond the scope of this paper.

18.1 Synopsis

Specific factors of sport climbing with relevance for pregnant women are the following:

- Experienced climber with uncomplicated pregnancy
- Risk of trauma
- Level of exertion
- Abdominal pressure applied by the sit harness vs. whole body harness
- Stage (first, second, or third trimester)
- Shift of the body's center of gravity during pregnancy
- Reduced stability of ligaments and joints

Principally, there is a need to weigh the advantages of continued sport participation during pregnancy against the possible disadvantages or risks. There are clear advantages to any woman who continues to exercise during her pregnancy: increase of psychological and physical well-being, better weight control, a reduction of gestational diabetes by almost 50% and a decrease of the risk of pre-eclampsia by about 40%, a lower incidence of preterm birth and Cesarean birth, and also an improvement of other factors, e.g., quality of sleep and less gestational depression [6, 10–18].

Three meta-analysis found that birth weights between those who exercised during pregnancy, or did not, was minimal to none. However, those who exercised “vigorously” during the third trimester were more likely to deliver infants weighing 200–400 g less than comparable controls, although there was not an increased risk of fetal growth restriction [2]. Vigorous was defined as up to 85% of capacity, though it is still not known what absolute level of intensity and duration, or both, that if exceeded could place the fetus at risk.

Trained women may continue their exercise during pregnancy and can realize significant strength gains [10, 19]. Table 18.1 provides an overview of the relative and absolute contraindications and warning signs which should be considered.

The risk of trauma is very low in sports climbing compared to other sports—e.g., the injury incidence is significantly lower than in contact sports or even school volleyball. Table 18.2 provides more detail. Most of the injuries sustained in sports climbing are of minor severity [20]. Prolonged medical treatment or in-patient treatment is extremely rare in relation to the large number of climbers [1, 21, 22]. The injury incidence in pregnant versus non-pregnant women was marginally higher, 0.28 versus 0.23/10.000 h of climbing, respectively, but all of these injuries were again of very minor severity, and such data are scarce [20]. The annual accident statistics of the German Alpine Club and others show that more than 50% of all severe accidents in sports climbing are caused by significant mistakes and failures during belaying [23]. This reinforces

Table 18.1 Relative and absolute contraindications for sport climbing during pregnancy and warning symptoms (Modified and amended from ACOG Committee Opinion, 2020)

Relative contraindication	Absolute contraindication	Warning symptoms
Severe anemia	Incomplete closure of the cervix	Vaginal bleeding
Extreme underweight (anorexia)	Persistent bleeding in the second or third trimester	Dizziness or sight disorder
Not clarified arrhythmia of the mother	Premature rupture of the membranes	Headache
Chronic bronchitis	Gestational hypertension	Thoracal pain, palpitations, conspicuous fatigue
Stress-induced asthma	Cardial disease with significant hemodynamic effect	Muscular weakness affecting balance
Heavy smoking	Restrictive pulmonary disease	Dyspnea before onset of exercise
Any of the following situations if not stabilized perfectly: Type 1 diabetes, hypertension, seizures, preeclampsia, hyperthyreosis	Multiple high-risk pregnancies or preterm birth	Edemas of the lower leg until deep vein thrombosis has been excluded
Orthopedic limitations	Placenta previa later than week 26	Reduced fetal activity
Couch potato	Premature uterine contractions	Premature uterine contractions or abdominal pain
Intrauterine retardation		Outflow of amniotic liquor
Patent foramen ovale with significant right-left shunt		Sudden onset of edemas of the ankle, face, or hands

the need that all climbers—pregnant or not—must always have regard for perfect belaying technique.

Table 18.2 Injury and trauma risk of several sport disciplines. Note: The table does not give any information about the severity of the respective injuries (Modified and amended from [20, 21, 24, 46–48])

Discipline	Accidents per 1000 h of activity	Discipline	Accidents per 1000 h of activity
Rugby, amateurs	283	Soccer National League Germany	3.10
Rugby, professionals	150	Competition climbing	3.10
Ice hockey, professionals	83	Triathlon	2.50
Rugby, youth	57	Boxing	2.00
Handball, women	50	Mountain biking	1.00
Soccer, champions league	31.6	Ski/Snowboard (downhill)	1.00
“Classic” mountaineering, 1980s	37.5	Nordic walking	0.90
Motorbike sport	22.4	“Classic” mountaineering, 2010s	0.56
American football	15.7	Wind surfing	0.41
Sailing	8.8	Sport climbing	0.027–0.079
Polo	7.8	Sport climbing, men	0.19
Kite surfing	7.0	Sport climbing, women	0.23
Volleyball at school	6.7	Sport climbing, pregnant women	0.28
Ice climbing	4.07		

Generally, it is recommended that pregnant women should avoid disciplines with increased risk for trauma. Compared to other sports, the risk of injuries in sports climbing is very low (Table 18.2). Additionally, it should be pointed out that most of the few injuries which occur are minor (e.g., abrasions or wounds), which do not require professional medical care with a score of NACA I or NACA II, and no fatalities [20, 24, 25]. However, there are some considerations which may reduce this risk further. First of all, climbing during pregnancy does not mean “pushing your limits” or competition climbing. During pregnancy, other aspects prevail as there are social factors and activity and exercise in general. Top rope belaying and avoiding the “sharp end” of the rope also increase the safety for the women although the height of a fall is usually low in sports climbing, and the bolts are perfectly fixed. Special care should be given to avoid “grounders” (falls to the ground), a type of accident which is still needlessly common in climbing gyms [23].

In general, the environment of an indoor climbing gym is safer than outdoor climbing where there are objective risks like breaking handholds, rock fall, scree, and others. However, other risks

are present at climbing gyms—crowded halls and mass gatherings especially in early evening or at weekends. Here, climbers should take care to be socially distanced to minimize collision accidents when a climber falls from above or nearby [23].

A specific problem for pregnant climbers concerns the risk of trauma by wearing a sport climbing harness. In the first trimester, the fetus is situated completely within the pelvis, and this protects it as a knight’s armor would do. Therefore, there is no problem with the fitting of the harness. During the second trimester, the abdomen will enlarge and emerges from the protective pelvis presenting two problems. The first concerns a shift forward of the center of gravity, a change in postural balance which will often go unnoticed and uncompensated for some long time by the women because it develops slowly. For example, falling is a common cause of injury in the general pregnant population as pregnant women are two to three times more likely to fall compared to non-pregnant women due to changes in center of gravity and joint laxity [10]. The climber’s growing bump and larger breasts will push her further away from the wall influencing both grip and footwork. Try climbing with more “open” positions rather than twisting,

deep bending, or crunch moves as the pregnancy progresses. The second problem that may arise is that the climbing harness may cause direct pressure on the lower abdomen, especially in the case of a fall, though there is no data. Nevertheless in the third trimester, the woman should consider switching to using a classic whole-body harness where the abdominal strap is located extremely low. Wearing one when abseiling or being lowered can feel awkward and “pushed out” as the woman’s center of balance will be moved as the rope is tied in at the chest and not the waist as with a sport climbing harness.

Some women develop edema, including to the feet, especially in the third trimester [10]. This may necessitate the need for new climbing shoes. Generally, comfortable climbing shoes which do not constrict are recommended [26]. Edema may be linked to pre-eclampsia and hypertension which should be investigated; however, there is no published data linking edema to prenatal exercise [10].

The normal course of pregnancy causes significant physical and metabolic load for the body which necessitates regular rests. Resting cardiac output increases by 50% [6, 27], and tidal volume and minute volume also increase significantly [6, 28]. Add to this a steadily increasing rise in the diaphragm’s position, and it is not surprising that many pregnant women complain about shortness of breath, especially when exercising [6, 29]. Unlike cycling or running where exercise programs can simply prescribe a steady state heart rate to aim for, this steady state goal has never been applied to sport climbing whether pregnant or not as it is short bursts of intermittent exercise. When feeling breathless or uncomfortable, climbers generally enjoy a break at an early stage of such complaints [26]. For any kind of sports or work, pregnant women should know that any subjective estimation of load by pulse rate is not reliable during pregnancy and that the load is significantly higher than the subjective estimation would be [6, 30].

The increase of body weight and changes on the angles of the joints may cause significant musculoskeletal adjustments and loads [6]. The stress of the knees and the hip joints may increase

by 100%. This may worsen preexisting instabilities or defects of these joints. Physiologically, the different hormonal pattern during pregnancy causes an increasing relaxation of the ligaments. Overload problems of the shoulders, ankles, and fingers in pregnant climbers were found [31, 32]. Some research found a general increased risk for injuries of the ligaments around the ankles in women, independent of pregnancy [20], while others did not [31]. Therefore, the influence of relaxed ligaments on the injury risk of pregnant women is somehow unclear. So far there is a lack of studies on sportive pregnant women [6]. Nevertheless, good climbing technique is recommended to avoid damages of ligaments and joints.

Choosing an appropriate climbing route is very important. First of all, to avoid “pushing the limits” means that the climb should be at least one grade easier (better two) than the woman was able to climb before her pregnancy. If she climbs routes which she already knows, this will further reduce the load and therefore the strain on her body. She should also not climb overhangs or roofs to avoid a significant increase of intra-abdominal pressure and favor slabs. High intra-abdominal pressure may decrease stability of the pelvic floor, and then problems like incontinence or prolapse may occur [6].

One of the rare studies concerning pregnant women climbers revealed some very interesting and relevant results, with the following figures reported as an average and the range in brackets [1]. The study group consisted of experienced climbers who had climbed for 11.1 years [2–21] before pregnancy, and 55.6% were normally leading. In the main, they continued climbing for 6.3 months of their pregnancy and were belaying for 5.8 months [2–9]. There was an inverse relationship whereby every 3 months the portion of women leading climbs decreased significantly, while top rope belaying increased. The climbing grade difficulty while leading decreased in the first trimester to 6.3 (4.00–9.33, UIAA decimal scale), to 5.8 (4.33–7.33) in the second trimester, and to 5.1 (4.33–7.33) in the third. The tendency was similar in top roping or seconding: 6.7 (4.33–8.0), 6.3 (4.00–7.66), and 5.6 (4.00–

7.66), respectively. The frequency of climbing also decreased from 32.2 h/month before pregnancy to 20.6 h in the first trimester, 16.3 h in the second, and 6.1 h in the third. These women did not receive any sport-specific advice, and yet the majority instinctively adapted their climbing exercise with their pregnancy to meet the same advice a doctor or midwife who is experienced climber would give. This also indicates that the acceptance of specific advice should be high in contrast to other situations in daily life. Those women who stopped climbing because of the pregnancy did so because of fear or unsureness concerning health of the fetus (33.9%), and in 7.1% they were pressed by their partner to do so. Both these reasons would indicate seeking professional medical advice to allay any concerns.

When the recommendations given above were followed, no negative consequences from the climbing activity on the fetus were found [1]. However, data are sparse. There is only one study available where this was investigated more in detail [33]. The study group consisted of 81% ($N = 32$) who were primipara (singleton pregnancy), and had a mean climbing experience of 9 years [2–24]. About half of them climbed until the third trimester, and 90% of them with reduced intensity. With a mean duration of pregnancy of 39.5 weeks (± 1.7), a mean birth weight of 3543gr (± 403) and a size of 50.9 cm (± 2.1) of exclusively healthy and completely developed babies were born after a normal duration of pregnancy. Two women gave birth at week 36, but there was no connection to their climbing activity.

Pregnant women should be made aware of environmental factors. It does not make any sense to climb in extreme heat [34]. They should also care for good fluid balance to optimize peripheral and placental perfusion.

Since there are sport climbing facilities not only at sea level but also at altitude in alpine villages, some notes are added here concerning altitude sojourns of pregnant women, especially if they do some exercise. Again we are only considering otherwise healthy women without a high-risk pregnancy here. Any recommendations are limited as most studies researched pregnant

indigenous populations who live permanently at high altitude (e.g., Tibetans) and whose generations of genetics will increase placental blood and oxygen supply to protect the fetus. Knowledge of pregnant lowlanders with acute exposure at moderate altitude (hours), exercising or not, is scarce (Leal et al., 2008).

We exclude the topic altitude diseases here for two reasons: (1) there are no gender-dependent differences concerning incidence, symptoms, prevention, and therapy of acute mountain sickness (AMS) [35] and (2) because pregnant women should prevent any altitude-related problem by adequate altitude profiles even more than non-pregnant people.

With respect to altitude medications and exotic locations, the use of acetazolamide is contraindicated during the first trimester (risk of teratogenicity) and after 36 weeks of pregnancy (risk of severe neonatal jaundice). Some drugs used for prophylaxis or treatment are also contraindicated during pregnancy: most antimalarials, quinolones, sulfonamide, and others [36].

Exercise presents competition for oxygenated blood supply to go to working muscles and the placenta, so there's a risk of fetal hypoxia or preterm labor [36]. Short stays without exercise, whether a few hours to a few days, are very low risk up to 2500 m, but no data is available. Long stays without exercise above 2500 m for weeks or months presents a higher maternal incidence of hypertension, pre-eclampsia, and placental abruption; and for the fetus there is intra-uterine growth retardation during the third trimester and low birth weight [36]. It is suspected there is a higher incidence of spontaneous abortion in the first trimester, but not proven [36]. The most commonly reported pregnancy complication by visitors to Colorado obstetrical care is preterm labor and bleeding complications [35]. Therefore, if the climber experienced difficulty becoming pregnant, it is recommended that she avoid high altitude.

There is no reason for any concern that the fetus may be in trouble if the woman goes to moderate altitude. For years it is well known that neurohumoral factors protect the fetus at rest and during exercise of the mother [37, 38]. An addi-

tional effect is caused by the hyperventilation of the mother which slightly overcompensates the effect of the reduced oxygen partial pressure at least up to 3100 m [39, 40]. Thousands of skiers prove these findings every winter. The fear that fetal growth retardation may occur has been proven for high altitude and prolonged stay only [41]. This can be neglected for sport climbing facilities in the European Alps.

The Medical Commission of the International Mountaineering and Climbing Federation (UIAA) gives conservative recommendations compared to the USA: acclimatized pregnant women may stay at 3000 m without any problem (USA: 4000 m). Significant exercise or workload should not be performed higher than 2500 m [36, 42, 43]. Other authors are even stricter—without supporting this by data—and recommend a limit of 1500–2000 m for significant workload [10, 44]. However, all such recommendations are fulfilled by sport climbing facilities in the European Alps and at most other places worldwide. There is a general consensus that above 2500 m any significant exercise should be started after 3–4 days of acclimatization only to avoid concurrency between the working muscles with the perfusion of the placenta [36, 41, 42, 45]. For the same reason, pregnant women should avoid hard exercise above 3000 m. It should be noted that these UIAA consensus recommendations were given by specialists from 35 countries which were established to their best knowledge because of a lack of good studies.

18.2 Conclusion

Extensive global literature strongly supports women with uncomplicated pregnancies to continue their aerobic and strength-conditioning exercises before, during, and after pregnancy, albeit with some modification due to anatomic and physiologic changes, type of exercise, and to fetal requirements. Experienced and active climbers with uncomplicated pregnancies can continue climbing throughout their pregnancy. Those who haven't climbed previously should

wait until after their pregnancy before beginning. Sport climbing overall has a low incidence of injury, with the majority of injuries minor in severity (NACA I or NACA II). The risk of trauma is very low in sports climbing compared to other sports (e.g., contact sports). Though data specifically on pregnant climbers is sparse, most pregnant climbers already instinctively adopt the recommendations presented in this paper, while others unnecessarily stop early as unsure about whether it is safe to continue as there is very limited data to inform this specifically. Sport climbing is also often incorrectly and subjectively perceived as being a high-risk activity by medical professionals who do not understand or have an appreciation of the sport and the ways to manage the risks. This paper presents those risks and how they can be modified. This includes selecting your routes carefully, avoiding climbing roofs and overhangs, not pushing your climbing grade, and considering switching to a whole body harness in the third trimester.

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